

UNSW-NB15 Data Dictionary

Machine Learning-Based Network Intrusion Detection

Formal public-facing reference for the variables used in the UNSW-NB15 network intrusion detection project.

Purpose

Formal reference document for the public project repository.

1. Dataset at a Glance

Observations	Columns	Training	Testing
257,673	45	175,341	82,332
Integer	Floating-point	Categorical	Candidate predictive
30	11	4	42

2. How to Use This Dictionary

The dictionary describes the 45 documented variables, their data types, their role in the modelling workflow and their meaning. The binary target is `label`. The identifier `id` is excluded from predictive modelling, while `attack_cat` is retained as an analysis variable and excluded from the predictive feature set.

3. Variables 1–15

#	Variable	Description / meaning	Type	Role
1	<code>id</code>	Unique identifier assigned to the network flow	<code>int64</code>	Identifier — excluded
2	<code>dur</code>	Duration of the network connection, measured in seconds	<code>float64</code>	Predictive feature
3	<code>proto</code>	Network protocol used by the connection	<code>object</code>	Predictive feature
4	<code>service</code>	Network service associated with the connection	<code>object</code>	Predictive feature
5	<code>state</code>	State of the network connection	<code>object</code>	Predictive feature
6	<code>spkts</code>	Number of packets transmitted from the source	<code>int64</code>	Predictive feature
7	<code>dpkts</code>	Number of packets transmitted to the destination	<code>int64</code>	Predictive feature
8	<code>sbytes</code>	Total number of bytes transmitted from the source	<code>int64</code>	Predictive feature
9	<code>dbytes</code>	Total number of bytes transmitted to the destination	<code>int64</code>	Predictive feature
10	<code>rate</code>	Data transmission rate during the network flow	<code>float64</code>	Predictive feature
11	<code>sttl</code>	Time To Live (TTL) value observed from the source	<code>int64</code>	Predictive feature
12	<code>dttl</code>	Time To Live (TTL) value observed from the destination	<code>int64</code>	Predictive feature
13	<code>sload</code>	Amount of data transmitted per second from the source	<code>float64</code>	Predictive feature

#	Variable	Description / meaning	Type	Role
14	dload	Amount of data transmitted per second to the destination	float64	Predictive feature
15	sloss	Number of packets lost from the source	int64	Predictive feature

4. Variables 16–30

#	Variable	Description / meaning	Type	Role
16	dloss	Number of packets lost to the destination	int64	Predictive feature
17	sinpkt	Average time between packets transmitted from the source	float64	Predictive feature
18	dinpkt	Average time between packets transmitted to the destination	float64	Predictive feature
19	sjit	Variation in packet timing from the source (source jitter)	float64	Predictive feature
20	djit	Variation in packet timing at the destination (destination jitter)	float64	Predictive feature
21	swin	Source TCP window size	int64	Predictive feature
22	stcpb	Source TCP base sequence number	int64	Predictive feature
23	dtcpb	Destination TCP base sequence number	int64	Predictive feature
24	dwin	Destination TCP window size	int64	Predictive feature
25	tcprtt	TCP round-trip time between source and destination	float64	Predictive feature
26	synack	Time between the SYN and SYN-ACK packets	float64	Predictive feature
27	ackdat	Time between the SYN-ACK and ACK packets	float64	Predictive feature
28	smean	Mean packet size transmitted from the source	int64	Predictive feature
29	dmean	Mean packet size transmitted to the destination	int64	Predictive feature
30	trans_depth	Depth of the connection within an HTTP transaction	int64	Predictive feature

5. Variables 31–45

#	Variable	Description / meaning	Type	Role
31	response_body_len	Length of the HTTP response body	int64	Predictive feature
32	ct_srv_src	Number of connections involving the same service and source	int64	Predictive feature
33	ct_state_ttl	Number of connections sharing the same state and TTL characteristics	int64	Predictive feature
34	ct_dst_ltm	Number of connections involving the same destination in the recent time window	int64	Predictive feature
35	ct_src_dport_ltm	Number of connections from the same source to the same destination port in the recent time window	int64	Predictive feature
36	ct_dst_sport_ltm	Number of connections to the same destination from the same source port in the recent time window	int64	Predictive feature
37	ct_dst_src_ltm	Number of connections between the same source and destination in the recent time window	int64	Predictive feature
38	is_ftp_login	Indicator of whether an FTP login was performed	int64	Predictive feature
39	ct_ftp_cmd	Number of FTP commands associated with the connection	int64	Predictive feature
40	ct_flw_http_mthd	Number of HTTP methods associated with the connection	int64	Predictive feature
41	ct_src_ltm	Number of connections involving the same source in the recent time window	int64	Predictive feature
42	ct_srv_dst	Number of connections involving the same service and destination	int64	Predictive feature
43	is_sm_ips_ports	Indicator of whether the source and destination IP address and port are the same	int64	Predictive feature
44	attack_cat	Category of attack associated with the network flow	object	Analysis variable — excluded
45	label	Binary target indicating normal (0) or malicious (1) traffic	int64	Target variable

6. Modelling Traceability

Variable / group	Role in the workflow
label	Binary target: 0 = normal traffic; 1 = malicious traffic.
id	Identifier; excluded from predictive modelling.
attack_cat	Analysis variable; excluded from predictive modelling.
proto, service, state	Categorical predictive variables requiring encoding.
Other candidate features	Subject to preprocessing, feature selection and model development.

7. GenAI-Assisted Document Generation

Generative AI was used as a controlled documentation and production assistant. The objective was to transform documented project information into a consistent public-facing reference without creating new dataset results or scientific conclusions.

Stage	Use of GenAI	Validation / boundary
Source review	Reviewed the documented dataset dimensions, variable names, types, roles and definitions.	Only documented project information was used.
Information extraction and structuring	Reorganised the variable information into a consistent dictionary format.	Variable count and modelling roles were preserved.
Editorial and layout	Improved wording, hierarchy, table design, pagination and visual consistency.	No new variables or modelling claims were added.
Format transformation	Prepared the same dictionary content for formal PDF and reusable spreadsheet formats.	The formats serve different documentation uses; the underlying content remains aligned.
Human validation	Checked terminology, values, variable roles and presentation before publication.	Final artefacts are subject to human review.

GenAI boundary. GenAI supported extraction, organisation, editorial improvement, layout and format transformation. It was not used to invent dataset results, modelling results or scientific conclusions.